

Special Course

Hard Carbon Synthesis & Characterisation for Sodium-Ion Anodes
(BioNa / DTU Energy / DTU Physics)

Project Information

- **Project owner:** BioNa (DTU startup), hosted jointly at DTU Energy & DTU Physics
- **Location:** DTU, Kongens Lyngby
- **Academic supervision:** Prof. Johan Hjelm (johh@dtu.dk) & Prof. Timothy Booth (tibo@dtu.dk)
- **Industrial supervision:** Dr. Peter Curran (pcurran@dtu.dk) & Szymon Dyszewski (szydys@dtu.dk)

Overview

This project gives the student hands-on experience in hard carbon production, structural characterisation, and half-cell assembly. The work supports BioNa's longer-term goal of optimising hard carbon anodes for sodium-ion batteries and moving material from lab scale toward scalable production.

Course Description

- Run an initial 11 pyrolysis experiments in 100 °C intervals: 600°C : 100°C : 1600°C to establish a baseline understanding of the carbonization process.
- Characterize the samples with Yield (mass-out / mass-in), X-Ray Diffraction, Raman spectroscopy, and N₂ - gas adsorption (BET fitting)
- Construct half-cells in coin-cell format and conduct Galvanostatic Charge/Discharge (GCD) screening to evaluate: Reversible Specific Capacity (mAh/g) & Initial Coulombic Efficiency (%)
- Use the Baseline carbonisation as a reference point to demonstrate a proof-of-principle for advanced "smart temperature profiles" by varying the ramp rate and introducing dwell points at intermediate temperatures. Ultimately defining a follow-on campaign of experiments.

Learning Objectives

Students completing the special course will be able to:

1. Safely operate high-temperature furnaces to run controlled pyrolysis processes
2. Perform and interpret characterisation of functional carbons
 - X-Ray Diffraction; Raman Spectroscopy; N₂ - gas adsorption (BET fitting)
3. Receive an introduction to advanced characterisation techniques of Small Angle X-Ray Scattering and He-pycnometry - opportunity to observe collection and interpretation of data
4. Search academic literature for comparison and to give context for results
5. Interpret measurements, storyboard them to form a coherent narrative and ultimately define follow-on campaign(s) of experiments.
6. Under supervision prepare a prototype slurry formulation to coat hard carbons into viable anode sheets.
7. Under supervision construct half-cells in coin format and conduct basic electrochemical testing.
8. Produce an extended written report documenting methods, results, and conclusions

Support and scope

The student undertaking this project will receive in-person support from BioNa employees to increase experimental throughput and remote supervision from senior researchers for analysis

and interpretation. Affords the candidate opportunity to flexibly dictate the direction of project towards their own interests e.g. Cell testing, material synthesis, material characterisation.

Deliverables

- Extended report
- Experiment logbook
- Raw characterisation data (XRD, Raman, BET)
- Physically labelled hard carbon samples handed to BioNa